

Module III

MODULE III

Physico–chemical treatment methods: Application of sedimentation, coagulation, flocculation, adsorption, chemical precipitation, ion exchange, reverse osmosis and electro-dialysis process. Biological treatment methods: Theory and principle, modification of activated sludge process, bio-tower, rotating biological contactors, stabilization pond, aerated lagoon, sequencing batch reactors, High rate anaerobic treatment: anaerobic contact process, anaerobic rotating biological contractors, Anaerobic Expanded/Fluidized bed reactors, Upflow anaerobic sludge blanket reactor.

SEWAGE TREATMENT

- Can be classified into preliminary primary, secondary and tertiary (advanced) treatment.
- Extent of treatment depends on purpose of treatment.
- Treatment before disposal requires only primary or in some cases both primary and secondary treatment- Effluent standards to be followed
- For reuse of treated water advanced treatment required

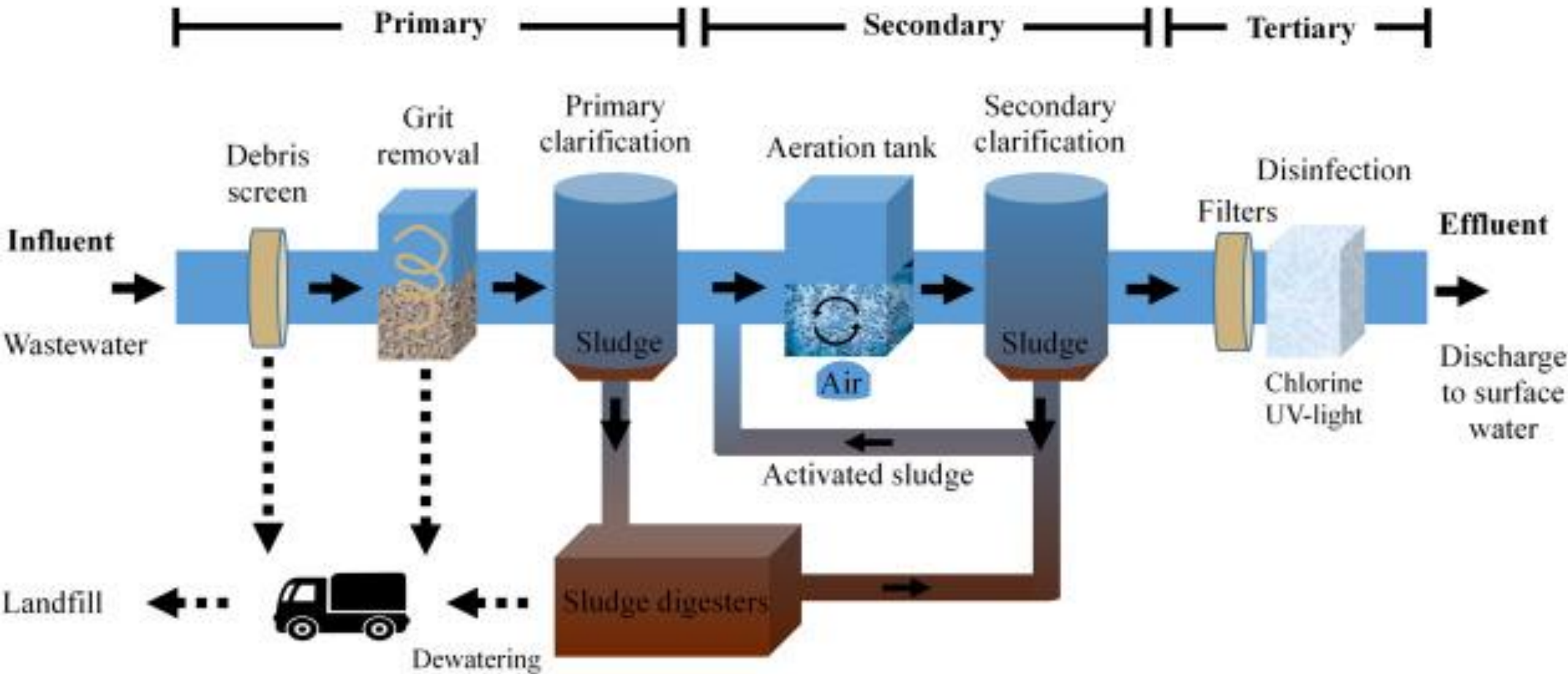
Objectives of wastewater treatment

- Wastewater treatment process and operations reduce objectionable properties of wastewater.
- - Removal of suspended and floatable matter.
- - Treatment of biodegradable organics
- - Elimination of pathogenic organisms
- - Removal of nitrogen, phosphorus, toxic organic compounds.

Steps in wastewater treatment

The Sewage Treatment Process





<https://www.youtube.com/watch?v=s8IVjQg7yno>

Steps in wastewater treatment

- **Preliminary Treatment** – for screening large floating material and removal of grits
- **Primary treatment** – the physical settling of insoluble solids from a wastewater stream, usually primary settling
- **Secondary treatment** – biological treatment system from the main wastewater stream
- **Tertiary treatment** – any treatment following secondary treatment, usually for the removal of nutrients, high TDS, microbial pollution etc.

Nagpur waste water treatment plant



Sedimentation/ Settling /Clarification

- Settle out the suspended materials due to gravity
- Suspended materials may be
 - Particles originally present in source water, such as clay or silts (Plain or primary settling)
 - The flocs created through coagulation-flocculation process (chemical added settling also called coagulant assisted settling)
 - Biomass produced in biological treatment units(secondary settling)

Settling is typically accomplished by decreasing the velocity of water to a point below which it no longer supports the transport of the particles, therefore gravity removes them from the flow

Sedimentation tank

- Tank allows suspended solids to settle under the influence of gravity
- Particles are settled at the bottom
- Settled particles are removed by scrapers
- Colour and turbidity will be removed

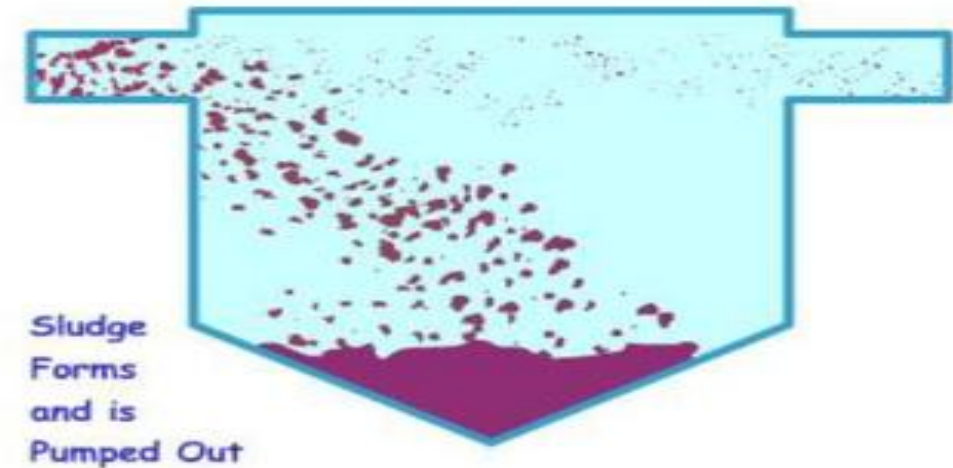
Classification of Sedimentation

- Type I Sedimentation
 - Characterized by the particles that settle directly at a constant settling velocity
 - They settle as individual particles and do not flocculate or stick to other during settling
 - Example: Sand and grit material
- Type II Sedimentation
 - Characterised by particles that flocculate during sedimentation
 - Due to their constantly changing size their settling velocity changes.
 - Example: alum or iron coagulation

Sedimentation Process

- Sedimentation is the process of removing solid particles heavier than water by gravity settling
- Oldest and most widely used unit operation in wastewater treatments
- Sedimentation basin may also be referred to as a sedimentation tank/clarifier/settling basin/settling tank

Sedimentation



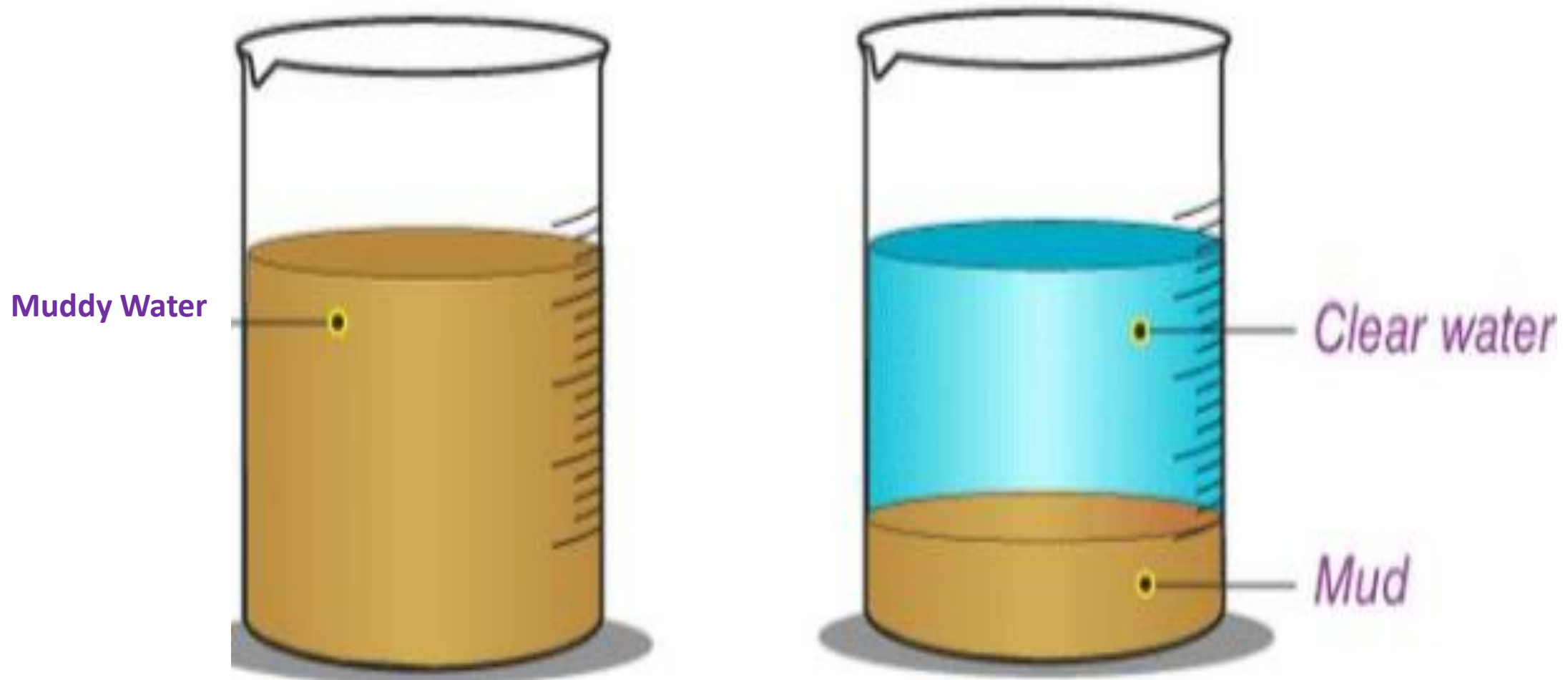
Sedimentation Process

- Process of allowing particles in suspension in water to settle out of the suspension in water to settle out of the suspension under the effect of gravity
- The particles that settle out from the suspension become sediment, and in water treatment is known as sludge.
- When a thick layer of sediment continues to settle , this is known as consolidation
- When consolidation of sediment , or sludge , is assisted by the mechanical means then this is known as thickening.

Sedimentation Principle

- Particles whose density is higher than that of the solvent sink and the particles that are lighter than it float to the top. Greater the difference in density, the faster they move
- Sedimentation may depend to the objects of various sizes ranging from large rocks in flowing water, to suspensions of dust and pollen particles, to cellular suspensions, to solutions of single molecules such as proteins and peptides
- Even small molecules supply a sufficiently strong force to produce significant sedimentation.

Sedimentation Process



Factors affecting sedimentation

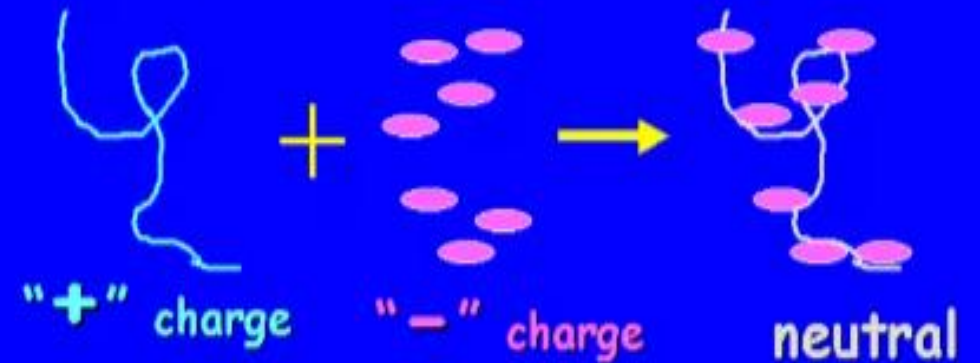
- Particle size, density and concentration and fluid viscosity etc affects the sedimentation rate
- Includes particle shape and orientation, convection currents in the surrounding fluid and chemical pretreatment of the feed suspension
- Alum increases the rate of sedimentation because alum being an external agent causes the particles present in the container to colloid together and form heavier particles, as heavier particles settle down therefore increasing the speed of sedimentation
- General increase in the rate of sedimentation with temperature. 20 C rise in temperature doubles the rate of sedimentation.

Coagulation

- Definition
 - Destabilisation of colloid particles by the addition of chemicals
- Application
 - Industrial waste containing colloidal and suspended solids

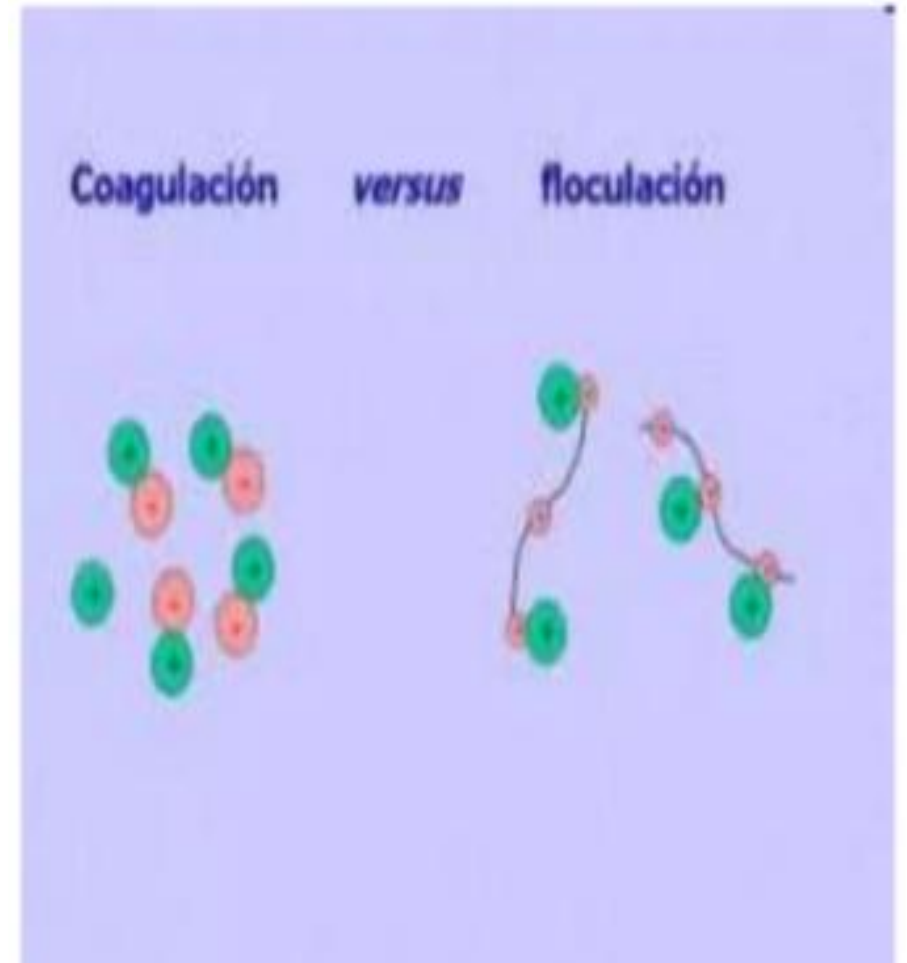
Purpose of Coagulant

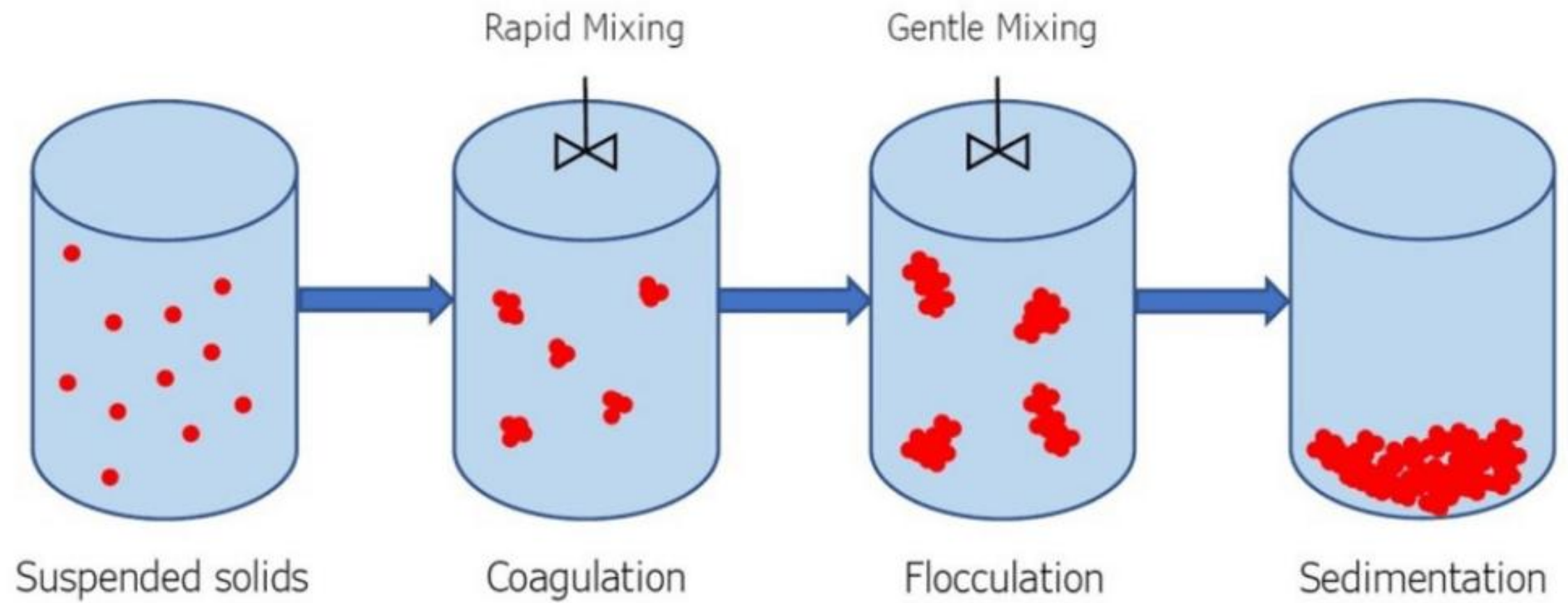
Neutralize excess charge

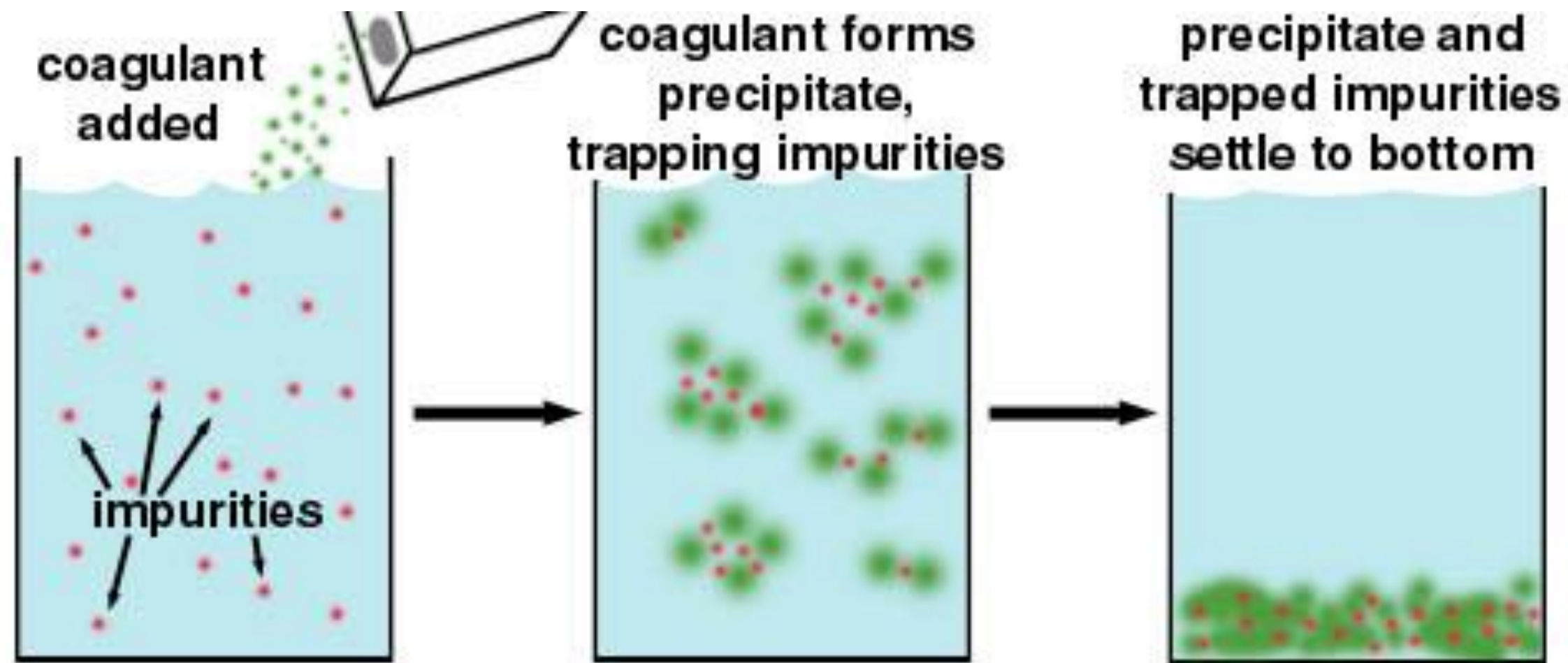


Flocculation

- Slow mixing technique which promotes the agglomeration of the stabilized particles







Factors affecting coagulation

- Types of coagulant
- Quantity or dose of coagulant
- Characteristics of wastewater
 - Type and quantity of suspended matter
 - Temperature
 - pH
- Time, violence and method of mixing

Coagulant

- Metal coagulant
 - Aluminum-based coagulants , ferro-based coagulants, magnesium chloride
- Organic polymer coagulants
 - Polyarylamide, Chitosan, Moringa Olifeira alginates

Coagulants contd...

- Other coagulants
 - Aluminium sulphate or Alum
 - Iron salts like ferrous sulphate/ferric chloride/ferric sulphate
 - These are suitable in slightly alkaline conditions
 - If not, lime or sodium carbonate are added
 - Sodium Aluminate
 - Aluminium Chloride
 - Chromic Chloride
 - Titanium Tetrachloride

Coagulant agent



Alum



Magnesium chloride



Polyacrylamide



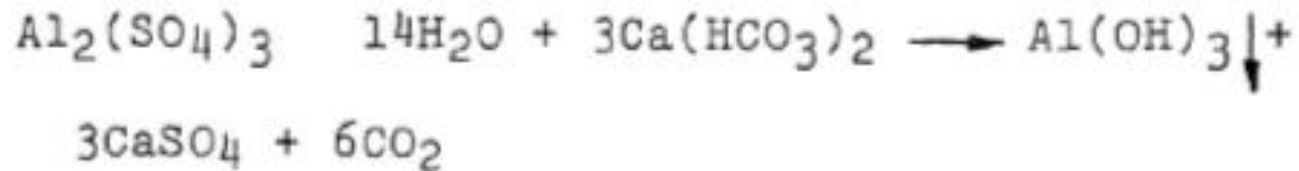
Moringa oleifera

Alum

- Most common, cheap & universal coagulant
- Requires the presence of alkalinity to form flocs
- Aluminium hydroxide formed is insoluble in wastewater



or



- Reacts with the water and results in positively charged ions

Advantages & Disadvantages of Alum

• Advantages

- It readily dissolves with water
- It does not cause the unsightly reddish-brown staining of floors, walls and equipment like ferric sulphate
- Skilled supervision not required

• Disadvantages

- It is effective only at certain pH range
- Good flocculation may not be possible with alum in some waters

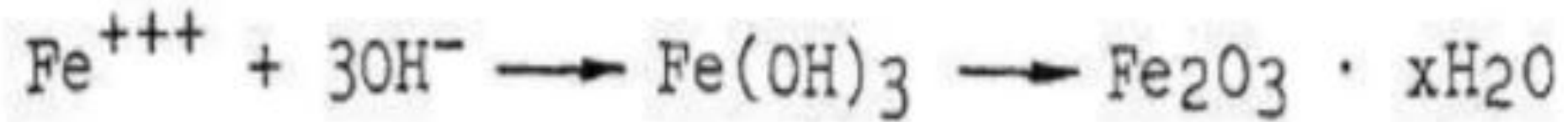
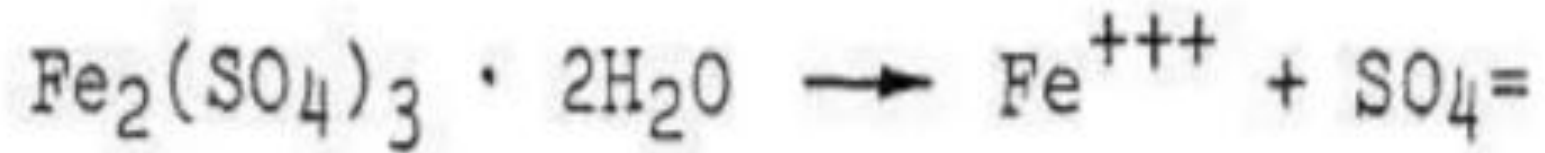
Ferrous sulphate

- Also known as copperas
- Usually added in conjunction with lime
- Reaction precipitates ferric hydroxide
- Suitable in the range of pH 8.5 and above

Ferric sulphate

- When Cl is added to a solution of copperas (ferrous sulphate) , ferric sulphate and ferric chloride forms
- The combination of ferric sulphate and ferric chloride --- chlorinated copperas
- Ferric sulphate and chlorinated copperas exhibit a dual range of effective pH, one from 4 to 6 and the other from 8 to 10
- Hence an effective coagulant for treating low and high pH

Ferric Sulphate



Advantages of ferric sulphate

- Ferric hydroxide is formed at low pH values so that coagulation is possible with ferric sulphate at pH values as low as 4
- Ferric hydroxide is insoluble over a wide range of pH values than aluminium hydroxide except for the zone of 7 to 8.5
- The floc formed with ferric coagulants is heavier than alum floc
- The ferric coagulants may be used in colour removal at the high pH values required for the removal of iron and manganese and in softening of water

Sodium Aluminate

- Sodium aluminates react with Ca and Mg of wastewater
- Resulting in the formation of precipitates of Calcium or Magnesium aluminate
- Costlier than alum

Comparison of alum and iron salts as coagulants

- Iron salts produce heavy floc and can remove much more suspended matter than alum
- Iron salts, being good oxidising agent, can remove hydrogen sulphide and its corresponding tastes and odours from water
- Iron salts can be used over a wider range of pH values
- Iron salts cause staining and promote the growth of iron bacteria in the distribution system
- Iron salts impart more corrosiveness to water than that which is imparted by alum
- The handling and storing of iron salts require more skill and control, as they are corrosive and deliquescent. Whereas, in alum no skilled supervision is required for handling alum

Merits of the coagulation process in wastewater treatment

- Sedimentation aided with coagulation produces better effluent with lesser BOD and suspended solids as compared to plain sedimentation
- A coagulated settling tank requires less space than that required by an ordinary plain sedimentation tank
- The coagulation process may also remove the phosphates from sewage, which may help in controlling the eutrophication of river, into which the effluent is being discharged

Demerits of the coagulation process in wastewater treatment

- More advanced methods of sewage treatment based on biological actions are available these days, and they are preferred to coagulation
- Chemicals used in coagulation react with sewage , and these reactions they destroy certain micro-organisms , which are helpful in digestion of the sludge, thus creating difficulties in sludge digestion.
- Cost of chemicals is added to the cost of sedimentation , without much use, and thereby making the treatment costlier

Demerits of the coagulation process in wastewater treatment contd...

- The process of coagulation and subsequent sedimentation produces larger quantities of sludge than that produced in plain sedimentation , and thus adding to the problems of sludge disposal
- The process of coagulation requires skilled supervision and handling of chemicals

Special cases

- For treating the wastewater from industries, using some specific chemical in their processes
- It is particularly advantages , where there is large seasonal variation in sewage flow or as an emergency measure to increase the capacity of an overloaded plain sedimentation tank. It may, thus be of use at places , like hill stations , where flow may considerably go up during rush seasons

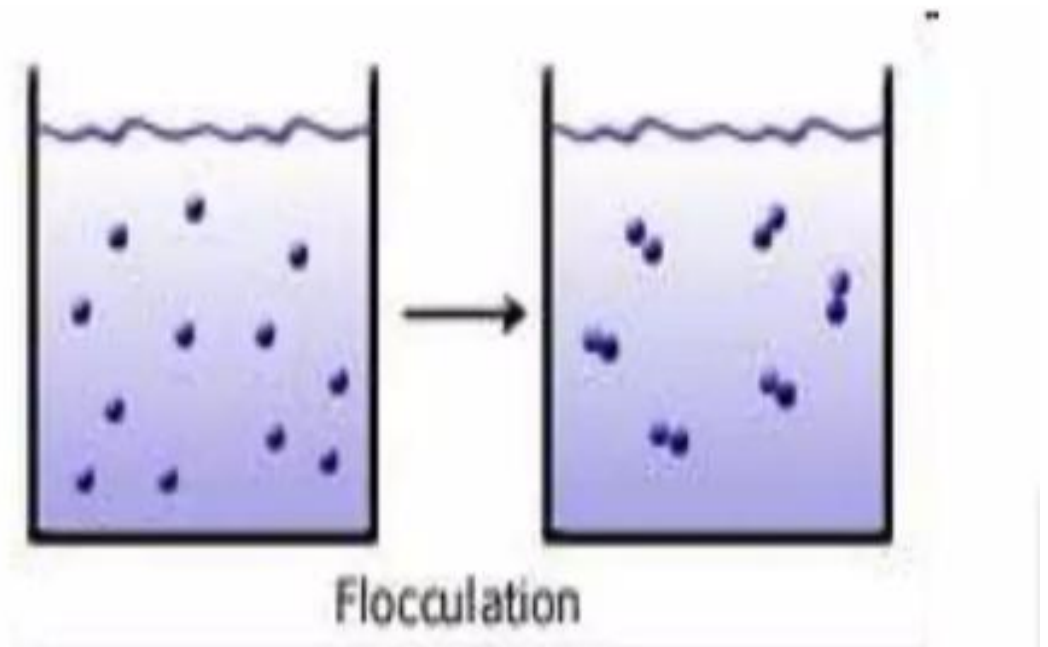
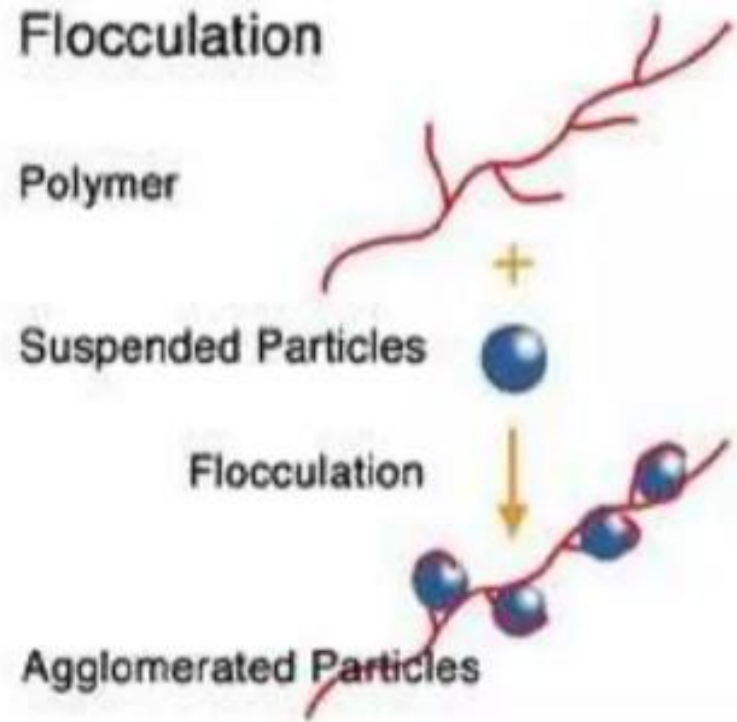
Coagulant Aids

- Accelerated the process of coagulation and flocculation by producing quick forming, dense and rapid-settling flocs
- Coagulant aids when added increase the density to slow settling flocs and toughness to the flocs so that they will not break up during the mixing and settling processes

Flocculation

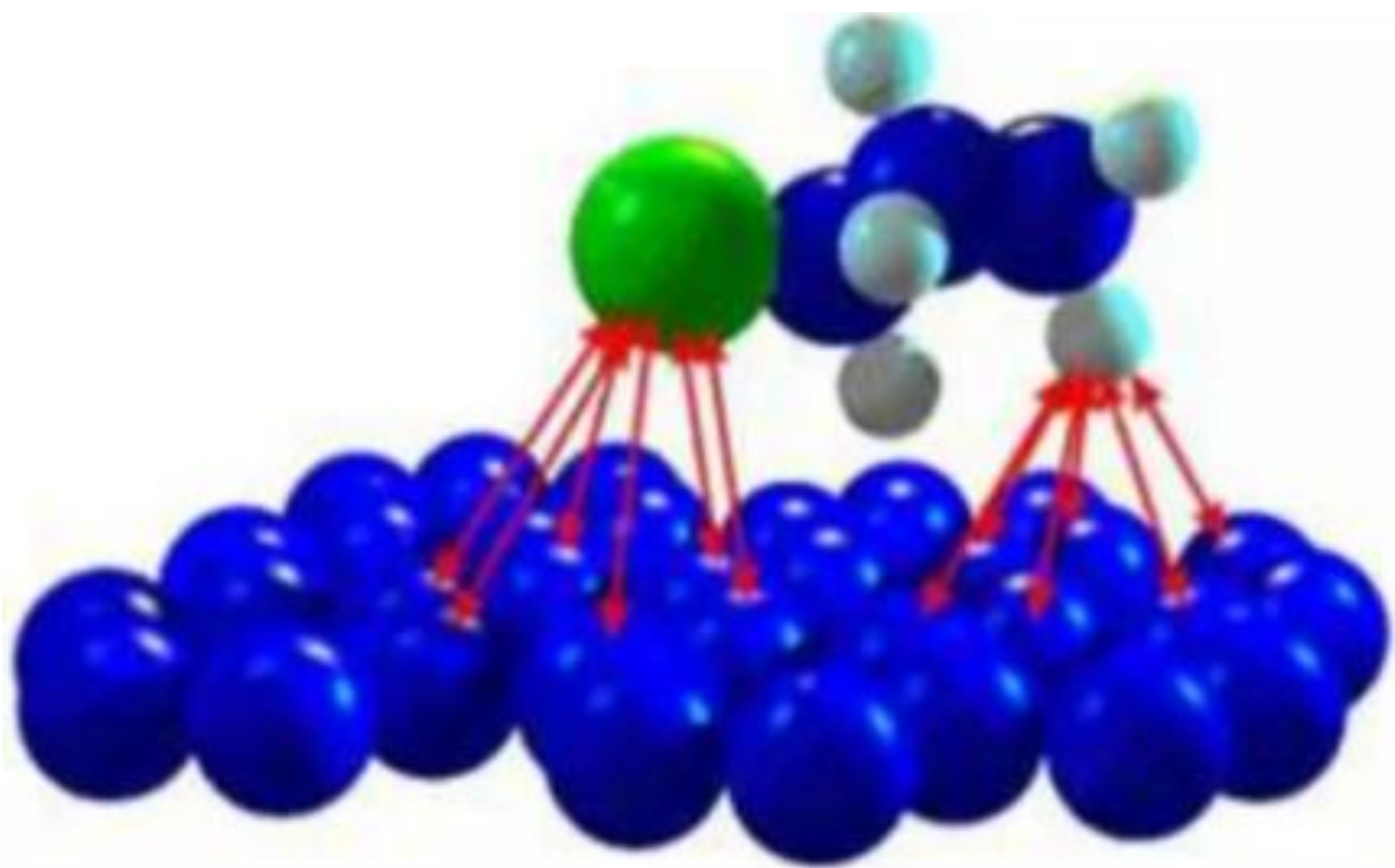
- Forming aggregate of flocs to form larger settleable particles
- Process includes
 - Mutual collision of small flocs resulting in bigger size
 - Usually slow speed or gentle mixing is used so as not to break the large flocs due to shear
 - Polymer or large molecular wt. compound is added to enhance flocs build up. Most of them are proprietary chemicals

Flocculation mechanism



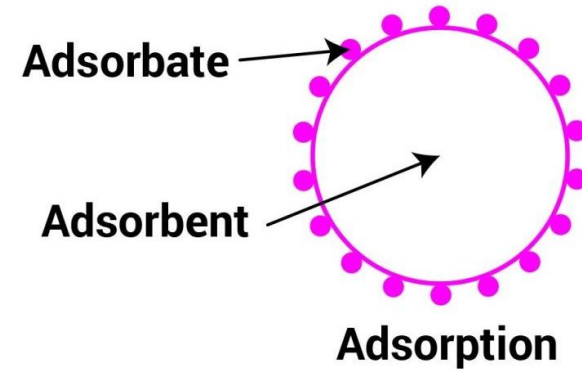
Effects of flocculation

- The benefits of flocculation are
 - To improve settling of particles in sedimentation tank
 - To increase removal of suspended solids and BOD
 - To improve performance of settling tanks



Adsorption

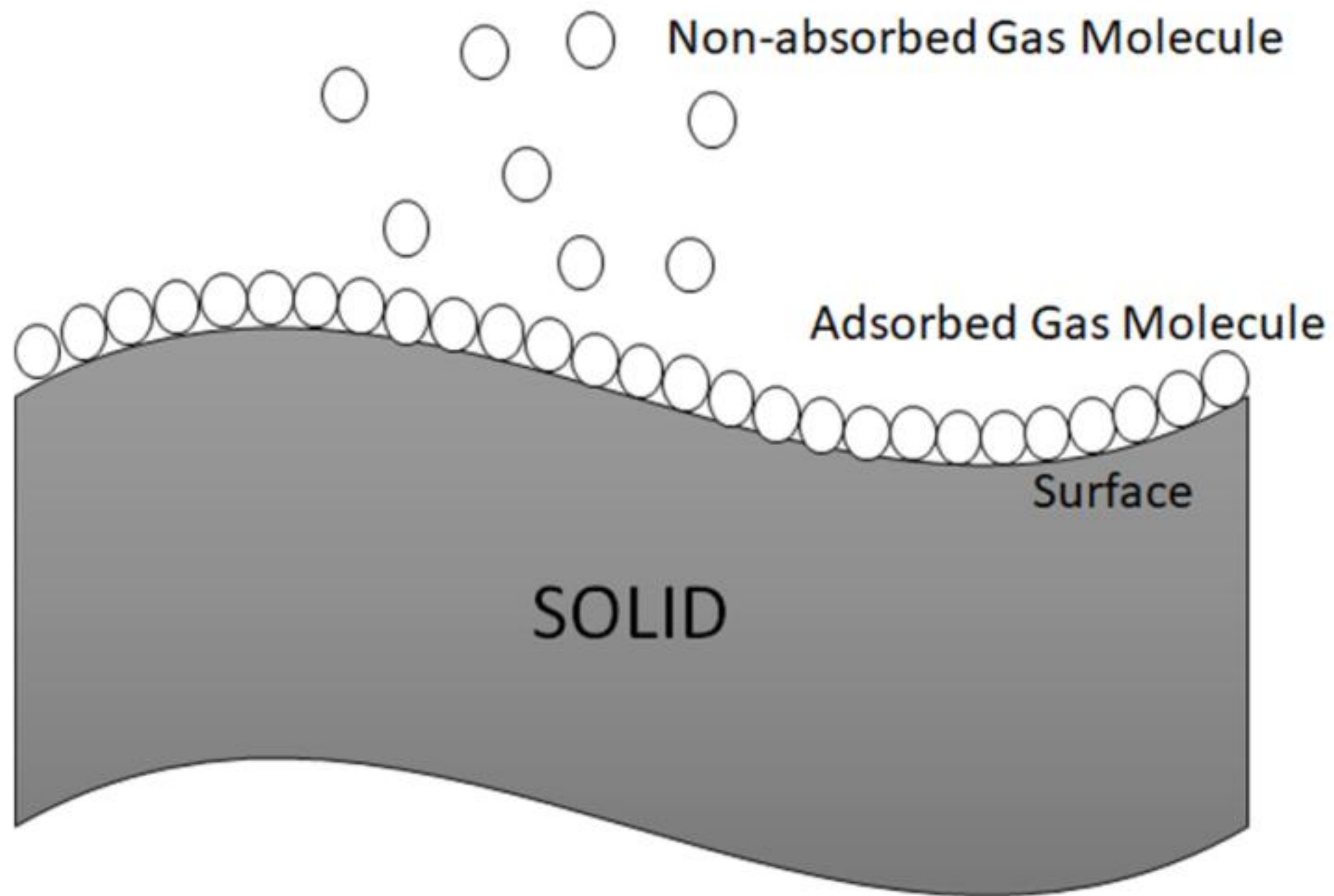
Adsorption



- Defined as a process in which atoms or molecules move from a bulk phase onto a solid or liquid surface
- In other words, it is adhesion of atoms, ions or molecules from a gas, liquid or dissolved solid to a surface
- Adsorbate – the substance which is adsorbed on the surface
- Adsorbent – the substance on which surface the adsorbate is adsorbed

Adsorption Principle

- Adsorption ---- Surface phenomenon & consequence of surface energy
- Surface of liquid or solid is in a state of strain which results in unbalanced residual force at the surface thereby resulting higher surface energy
- The surface of liquids/solids always have a tendency to attract & retain the molecular species with which it comes in contact
- This phenomenon is known as adsorption



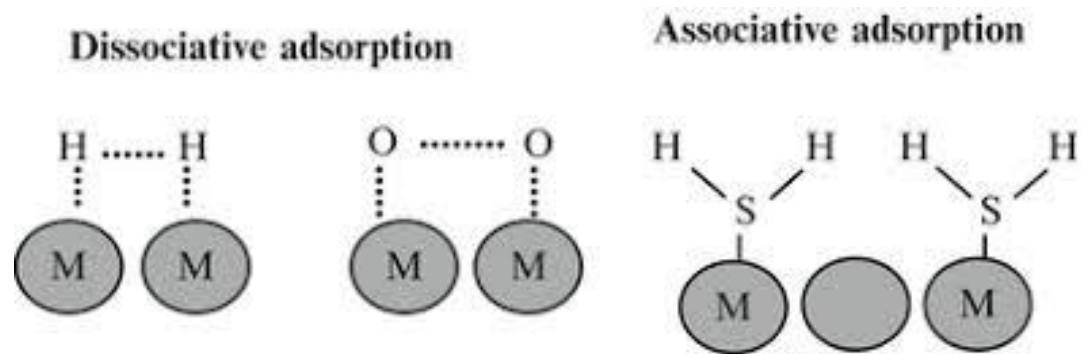
Types of adsorption

Type of adsorption

- Physical adsorption (physisorption)
- Associative chemical Adsorption (Chemisorption)
- Dissociative chemical Adsorption (chemisorption)

Characteristics

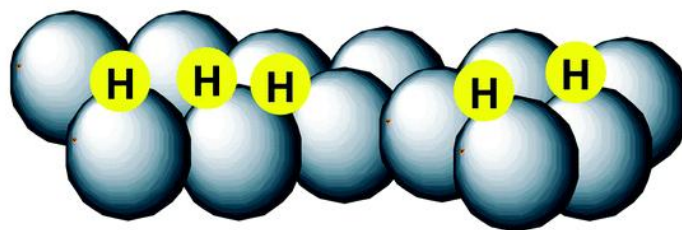
- Unselective. Low energy of adsorption
- Selective, strongly dependent on both gas & solid surface. Higher energies of adsorption than those of physisorption
- Selective, strongly dependent on both gas & solid surface. Higher energies of adsorption than those of physisorption



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*dissociative
exchange*

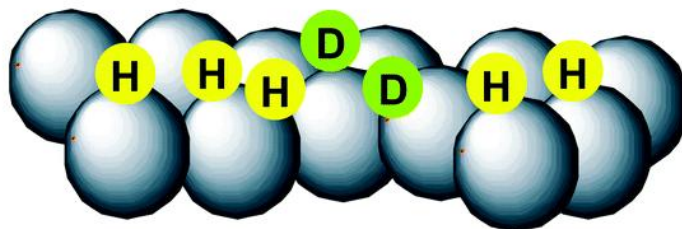


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exchange*

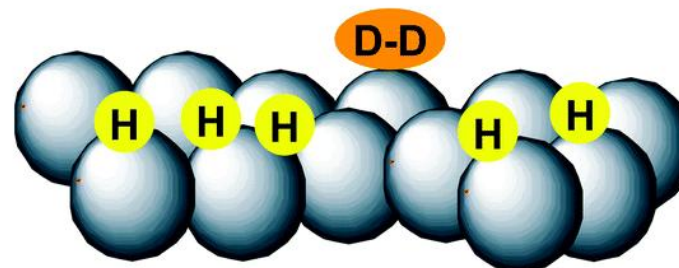
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adsorption* ↓

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D-D

Factors affecting adsorption

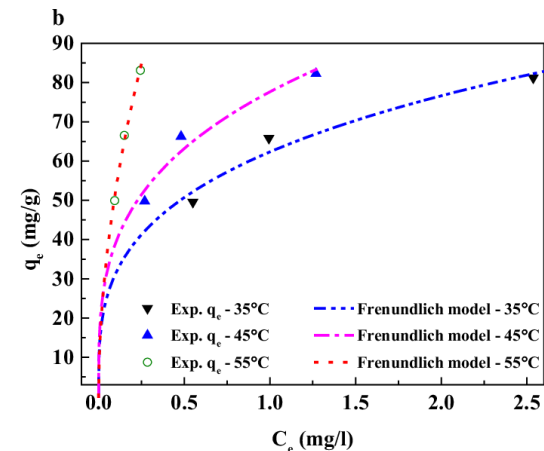
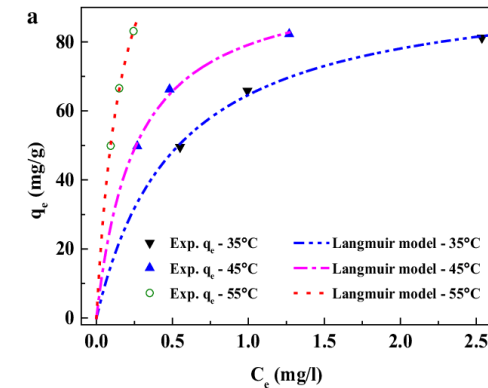
- Temperature
- Pressure
- Surface area
- Activation of solid adsorbent
- Agitation
- pH
- Characteristics of adsorbent
- Dose of absorbate & adsorbent

List of commercial adsorbents

- Activated Carbon
- Silica Gel
- Activated Alumina
- Synthetic Zeolite
- Clay
- Polymers and resins

Adsorption isotherm and its types

- Adsorption isotherm is a mathematical model or functional expression to depict the distribution of solute between the solid and liquid phase at equilibrium at a constant fixed temperature
- Types of adsorption isotherms
 - Langmuir isotherm
 - Freundlich isotherm



Applications of adsorption

- Effective in heterogeneous catalysis
- Effective in removing colouring material
- Effective as ion exchange resins
- Used as adsorption indicators
- Used in gas masks
- Used in dyeing of cloth
- Used as de-humidifiers

Scope of adsorption

- Adsorption is now recognized as an effective and economic method for removal of heavy metal & organic pollutants from water & wastewater treatment
- The adsorption process offers flexibility in design and operation and in many cases will produce high-quality treated effluent. In addition, because adsorption is sometimes reversible, adsorbents can be regenerated by suitable desorption process
- Certain techniques like coagulation, filtration with coagulation, precipitation, ozonation, adsorption, ion exchange, reverse osmosis and advanced oxidation processes have been used for the removal of organic pollutants from polluted water and wastewater
- These methods have been found to be limited , since they often involve high capital and operational costs.

Contd....

- Adsorption has advantages over the other methods because of its simple design and can involve low investment in terms of both initial cost and land required
- The adsorption process is widely used for treatment of industrial wastewater from organic and inorganic pollutants and meets the great attention from the researchers

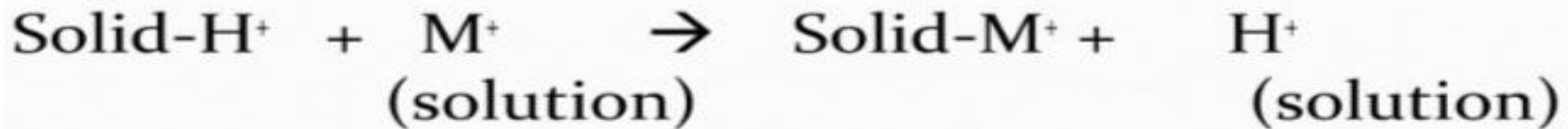
ION EXCHANGE PROCESS

- Process by which a mixture of similar charged ions can be separated by using an ion exchange resin
- Ion exchange resin exchanges ions according to their relative affinities
- There is a reversible exchange of similar charged ions
- Mostly similar charged ions like cations or anions can be separated by this technique

Principle

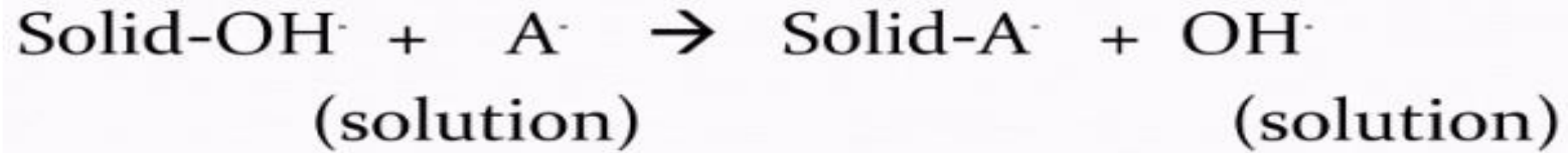
- Reversible exchange of ions between the ions present in the solution and those present in the ion exchange resin
- Cation exchange
- The cations retained by the solid matrix of ion exchange resin can be eluted by using buffers of different strength and hence separation of cations can be effected

Separation of cations

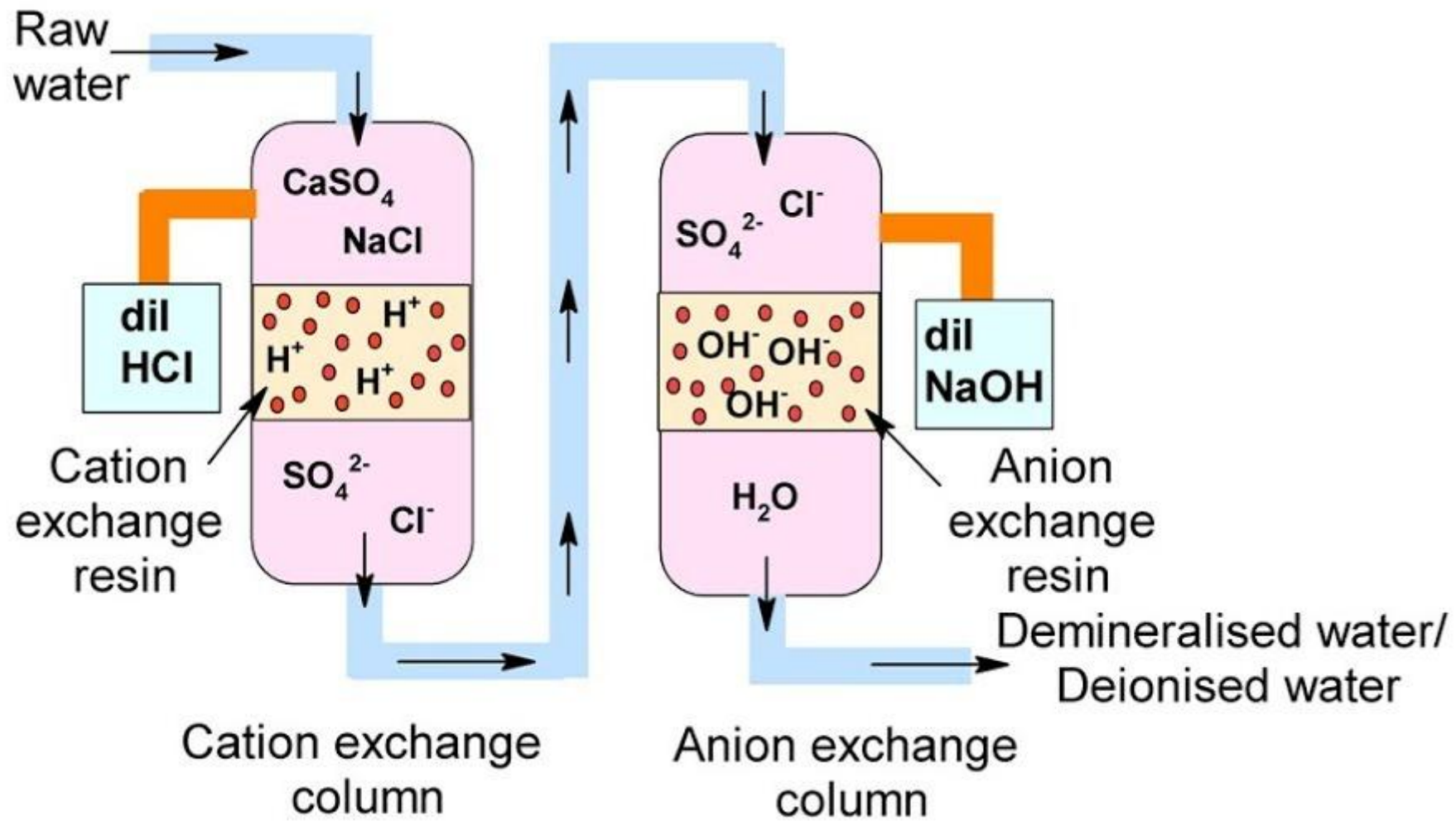


Anion exchange

- Separation of anions using
- Anion exchange resin



- The anions retained by the solid matrix of ion exchange resin can be eluted by using buffers of different strength



Classification of resins

- **According to the chemical nature**

- Strong cation exchange resin
- Weak cation exchange resin
- Strong anion exchange resin
- Weak anion exchange resin

- **According to source**

- Natural cation – zeolites, clay etc. natural anion –dolomite
- Synthetic – inorganic and organic resins
 - ❖ Organic resins are the most widely used
 - ❖ Organic ion exchange resins are polymeric resin matrix containing exchange

Functional group present in different ion exchange resins

- Strong cation exchange resin $\rightarrow \text{SO}_3\text{H}$
- Weak cation exchange resin $\rightarrow \text{COOH}, \text{OH}, \text{SH}, \text{PO}_3\text{H}_2$
- Strong anion exchange resin $\rightarrow \text{N}^+\text{R}_3, \text{NR}_2$
- Weak anion exchange resin $\rightarrow \text{NHR}, \text{NH}_2$

Common ion exchange resins

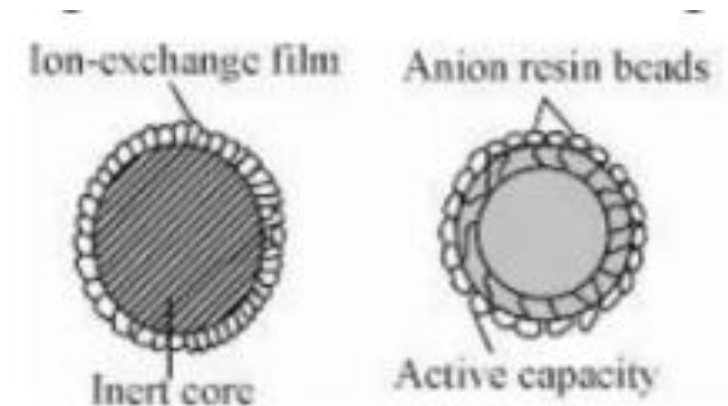
Class of resin	Nature	pH range	Applications
Cation –strong	Sulfonated polystyrene	1-14	Fractionation of cations,inorganic separations, peptides, aminoacids, B vits
Cation – Weak	Carboxylic methacrylate	5-14	Fractionation of cations, biochemical separations, org bases,antibiotics

Structural types of ion exchange resins

- ❑ Pellicular type with ion exchange film
- ❑ Porous resin coated with exchange beads
- ❑ Macroreticular resin bead
- ❑ Surface sulfonated and bonded electrostatically with anion exchanger

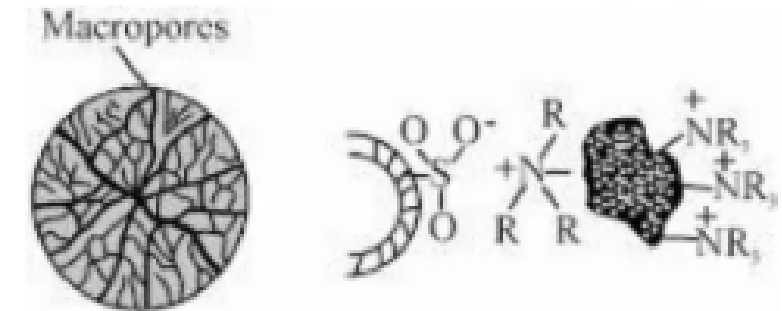
Structural types of ion exchange resins

- Pellicular type with ion exchange film
 - The particles have a size of 30-40 μ with 1-2 μ film thickness
 - These have very low exchange capacity
 - Ion exchange efficiency ; 0.01-0.1 meq/g of ion exchange resin
- Porous resin coated with exchanger beads
 - Size 5-10 μ
 - They are totally porous & highly efficient
 - Exchange capacities 0.5-2 meq/g or ion exchange



Structural types of ion exchange resins contd...

- Macrorecticular resin bead
 - A reticular network of the resin seen superficially on the resin beads
 - They are not highly efficient & have very low exchange capacities
- Surface sulfonated and bonded electrostatically with anion exchanger
 - The particles are sulfonated & they are bonded electrostatically with anion exchanger resin
 - They are less efficient & have low exchange capacity
 - Exchange capacity is 0.02 meq/g of exchange resin



Physical properties of resins

1. Particle size- 50-200 mesh fine powder

- Should allow free flow of mobile phase, should contain more exchangeable functional group

2. Cross linking & swelling

- When more cross linking agent is present, they are more rigid, but swells less
- When swelling is less, separation of ions of diff sizes is difficult as they can't pass through the pores present

Physical properties of resins

- When less cross linking agent is present, they are less rigid, but swell more
- Separation will not be efficient as exchange of functional groups does not take place due to wide pore
- Hence an optimum quantity of cross linking agent should be added to the polymeric ion exchange resin for the separation to be effective

Selection of ion exchange resins

- Depends upon
 - Type of ions to be separated – cations/anions
 - Nature of ions to be separated – strong/weak
 - Efficiency of resin- measured by ion exchange capacity
 - Ion exchange capacity is the total ion exchange capacity in terms of the exchangeable functional groups expressed as meq/g of the ion exchange resin
 - $\text{Meq/g} = 1000 \text{ eq/wt}$
 - Particle size of the resin – 50-100 /100-200 mesh
 - Structural type of the resin- porous, pellicular etc
 - Amount of cross-linking agent present – which decides swelling of the resin

Packing of the column

- Wet packing
 - Resin is mixed with the mobile phase and packed in the column uniformly
 - The sample to be separated is dissolved in the mobile phase and introduced all at once into the column

Mobile phase

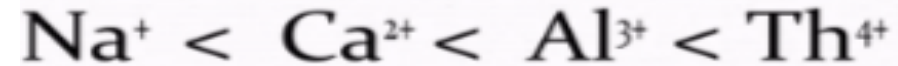
- Organic solvents are less useful and they are not used at all
- Only different strengths of acids, alkalies and buffers are used as eluting solvents
- Eg: 0.1N HCl, 1N NaOH, phosphate buffer acetate buffer, borate buffer, phthalate buffer etc

Regeneration of the ion exchange resin

- Ion exchange resin after separation may not be useful for next separation as exchangeable functional groups are lost
- But due to cost of the ion exchange resins they cannot be disposed off
- Hence reactivation, regeneration
- Regeneration makes the used ion exchange resin to be as efficient as a virgin resin
- Regeneration: replacement of the exchangeable cations or anions present in the original resin
- Hence regeneration of the cation exchange resin is done by the charging the column with strong acid like HCl

Factors affecting ion exchange separations

- Nature & properties of ion exchange resins- cross-linking & swelling
- Nature of exchanging ions –
 - **Valency of ions**- at low conc & ordinary temp, extent of exchange increases with increase in valency



- **Size of ions** – for similar charged ions, exchange increases with decrease in the size of hydrated ion $\text{Li} < \text{H}^+ < \text{Na}^+ < \text{NH}_4^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$
- **Polarizability** – exchange is preferred for greater polarizable ion
Eg: $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$
- **Conc of solution** – in dilute solutions, polyvalent anions are generally adsorbed preferentially

Factors affecting ion exchange separations

- **Conc and charge of ions** – if resin has higher +ve charge and solution has lower +ve charge, exchange is favoured at higher conc.
- If resin has lower +ve charge and solution has high +ve charge, then exchange is favoured at low conc.

Applications

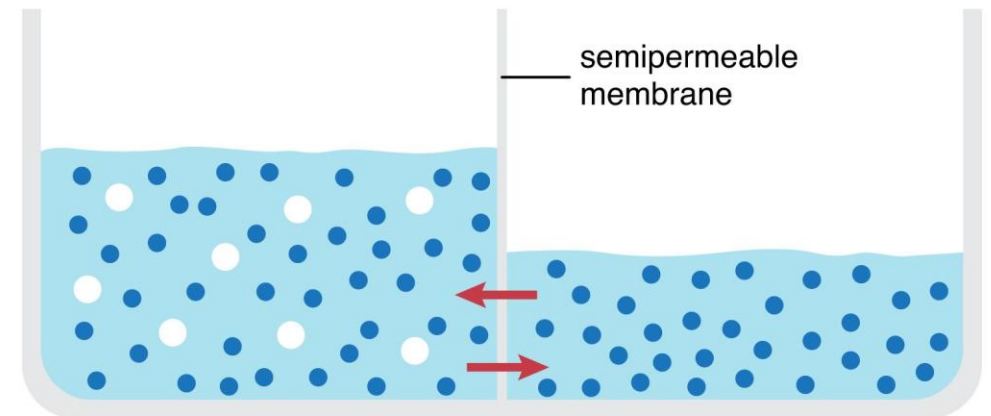
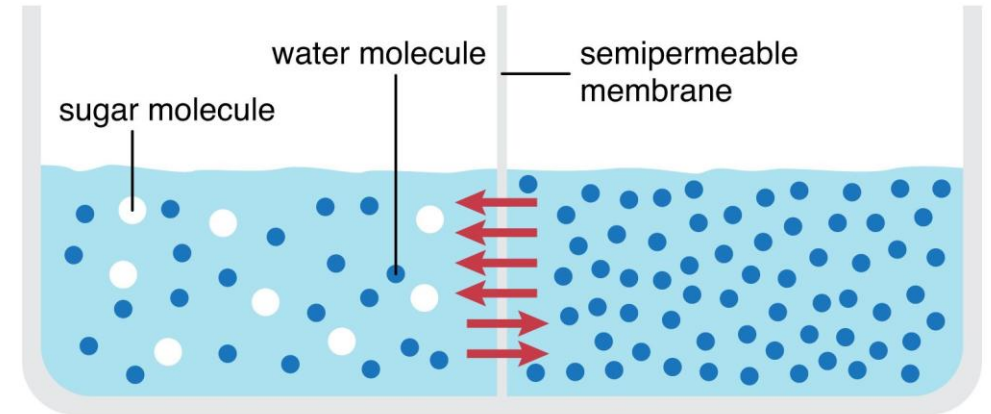
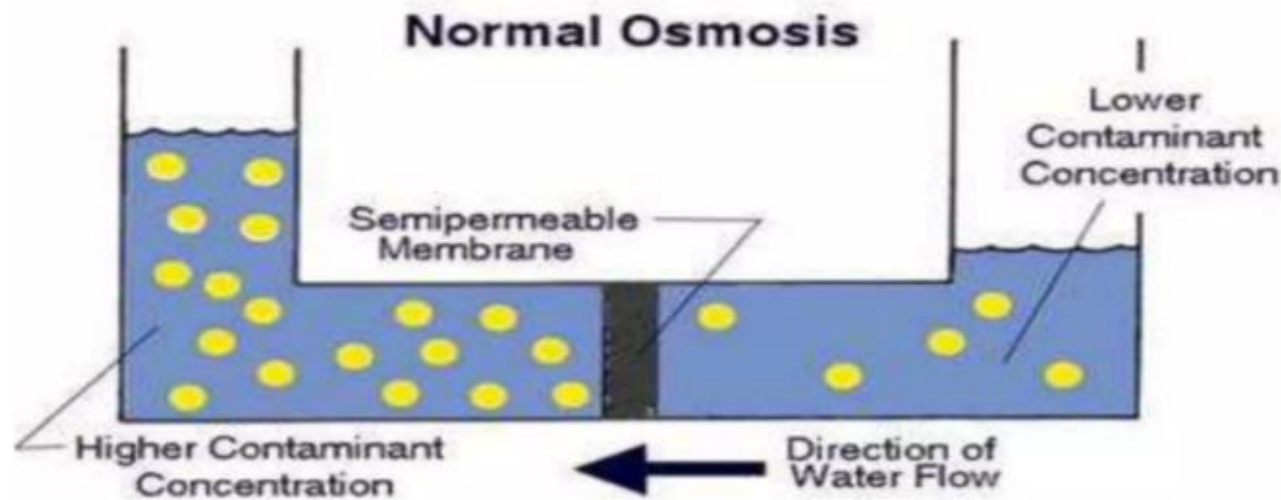
- Softening of water
- Demineralisation or deionisation of water
- Purification of some solutions to be free from ionic impurities
- Separation of inorganic ions
- Organic separations: a mixture of pharmaceutical compounds can be separated
- Biochemical separations like isolation of drugs or metabolites from blood, urine etc
- Conc of ionic solutions

Osmosis

- Osmosis is a naturally occurring phenomenon and one of the most important processes in nature
- It is a process where a weaker saline solution will tend to migrate to a strong saline solution

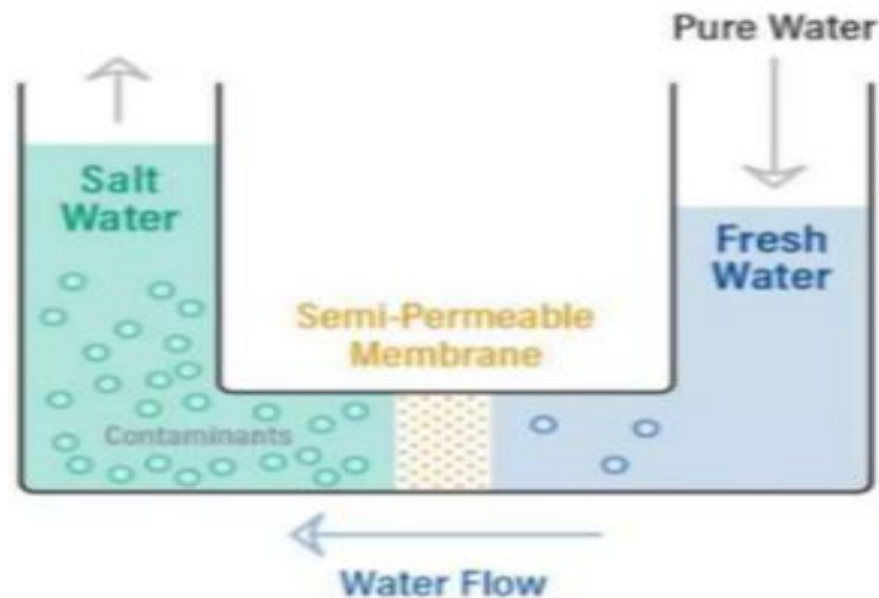
Definition of osmosis

- The solvent naturally moves from an area of low solute concentration to an area of high solute concentration called osmosis



Osmosis

- A solute that is less concentrated will have a natural tendency to migrate to a solution with a higher concentration
- A semi-permeable membrane is a membrane that will allow some atoms or molecules to pass but not others. A simple example is a screen door. It allows air molecules to pass through but not pests or anything larger than the holes in the screen door



This is a diagram which shows how osmosis works.

Reverse Osmosis

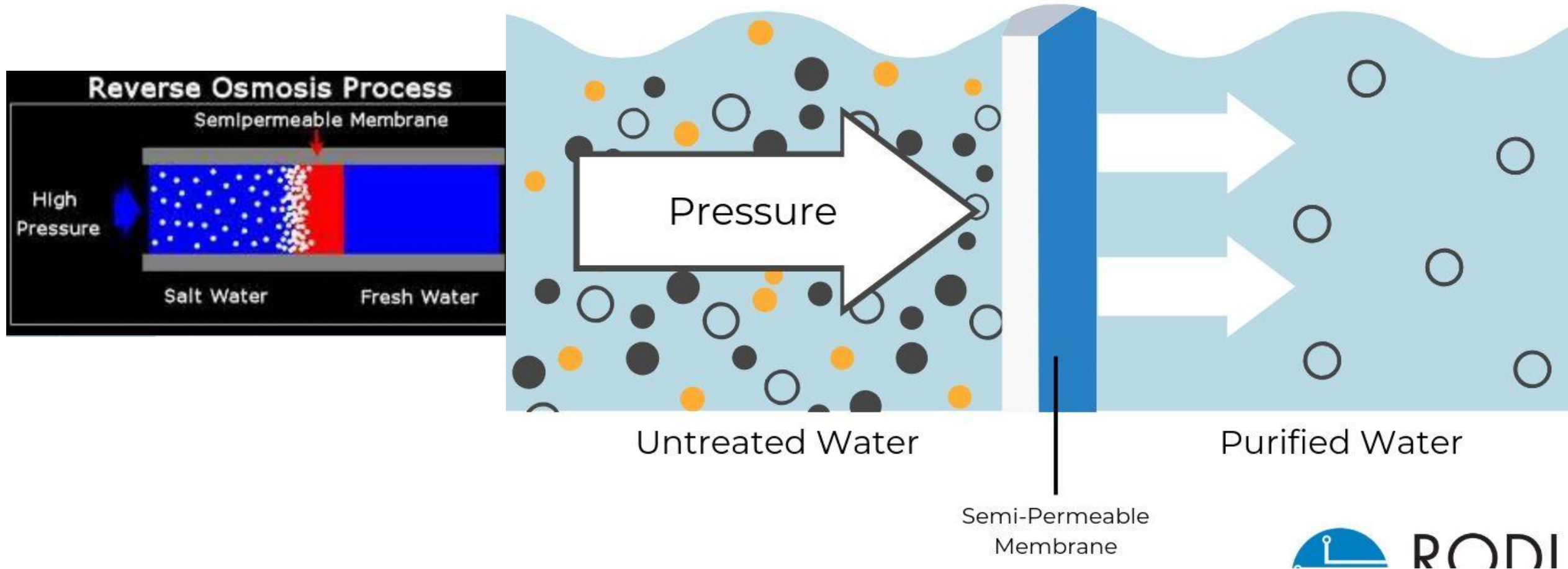
- Understanding reverse osmosis

Reverse osmosis, commonly referred to as RO, is a process where you demineralise or deionize water by pushing it under pressure through a semi-permeable reverse osmosis membrane

Intro to RO

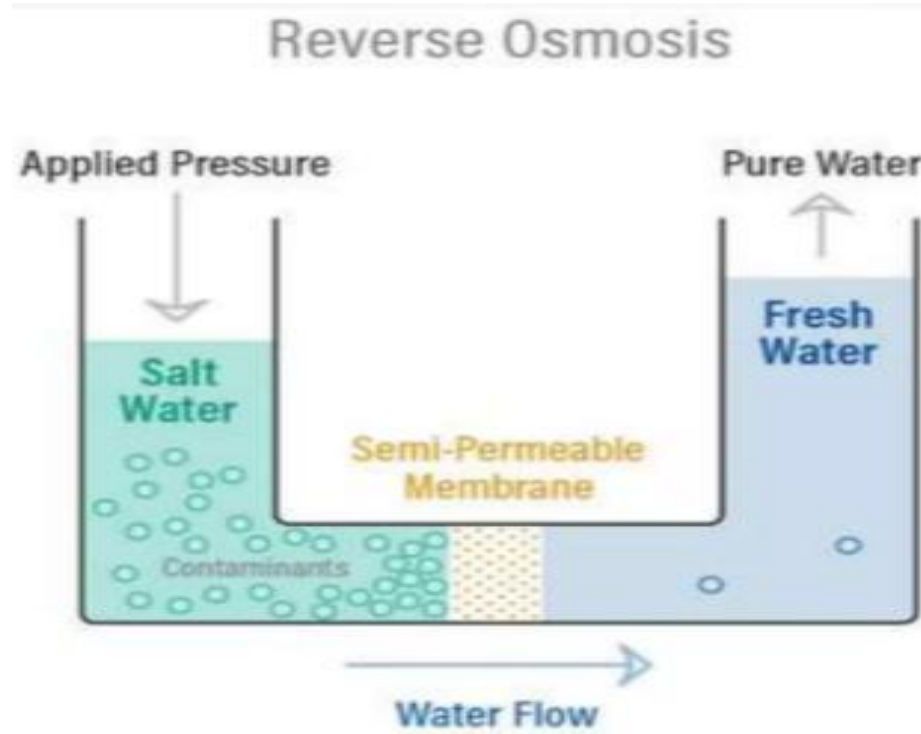
- RO is a water purification technology that uses semi permeable membrane
- It works on the concept of diffusion
- RO is also known as hyperfiltration
- RO can remove many types of molecules and ions from solution

Reverse Osmosis



RO contd....

- When pressure is applied to the concentrated solution, the water molecules are forced through the semi-permeable membrane and the contaminants are not allowed through



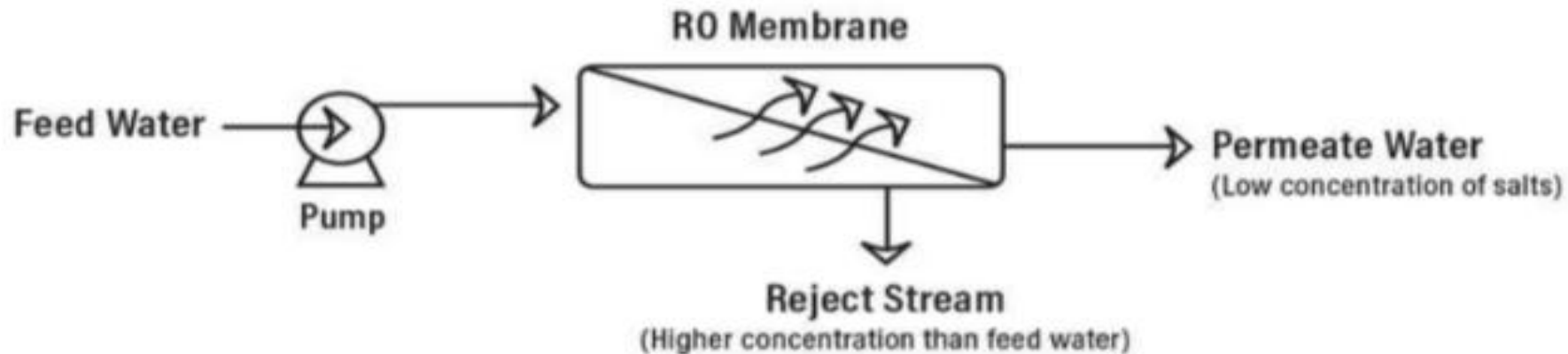
This is a diagram outlining the process of Reverse Osmosis.

RO is the process of osmosis in reverse

- Osmosis occurs naturally without energy required, to reverse the process of osmosis you need to apply energy to the more saline solution
- A reverse osmosis membrane is a semi-permeable membrane that allows the passage of water molecules but not the majority of dissolved salts, organics, bacteria and pyrogens
- However, you need to ‘push’ the water through the reverse osmosis membrane by applying pressure that is greater than the naturally occurring osmotic pressure

How does reverse osmosis work?

- Reverse osmosis works by using a high pressure pump to increase the pressure on the salt side of the RO and force the water across the semi-permeable RO membrane, leaving almost all (around 95% to 99%) of dissolved salts behind in the reject stream



How does reverse osmosis work? Contd....

- The desalinated water that is demineralised or deionised , is called permeate (or product) water
- The water stream that carries the concentrated contaminants that did not pass through the RO membrane is called the reject (or concentrate) stream
- It is important to understand that an RO system employs cross-filtration rather than standard filtration where the contaminants are collected within the filter media
- With cross filtration, the solution passes through the filter , or crosses the filter, with two outlets:

The filtered water goes one way and the contaminated water goes another way

What contaminants will RO remove from water?

- RO is capable of removing up to 99%+ of the dissolved salts (ions), particles, colloids, organics, bacteria and pyrogens from the feed water
- An RO membrane rejects contaminants based on their size and charge
- Any contaminant that has molecular weight greater than 200 is likely rejected by a properly running RO system

What contaminants will RO remove from water? Contd....

- Likewise, the greater the ionic charge of the contaminant, the more likely it will be unable to pass through the RO membrane, this is why an RO system does not remove gases such as CO₂ very well because they are not highly ionized and have a very low molecular weight
- RO is very effective in treating brackish, surface and ground water for both large and small flows applications

Dependent on the following factors

- Influent solute concentration
- Water flux rate
- Pressure
 - Pressure caused by the difference in solute concentration between the two compartments
 - The externally applied pressure
- Required pressure amount (highly concentrated side)
 - For fresh and brackish water: 15.5 to 26 bar
 - For seawater: 55 to 81.5 bar

Desalination

- Useful for areas that have either no or limited surface water or ground water
- RO is the most common method of desalination, although 85 percentage of desalinated water is produced in multistage flash plants
- Sea water RO (SWRO): No heating or phase changes are needed, energy requirements are low in comparison to other processes of desalination, but are still much higher than those required for other forms of water supply
- The Ashkelon SWRO desalination plant in Israel is the largest in the world

Applications

- Desalination of brackish water
- Desalination of seawater
- Production of ultrapure water
- Wastewater treatment

Use of RO technology

- Drinking water purification
- Wastewater purification
- Hydrogen production
- Maple syrup production in food industry
- RO used for production of deionised water

RO disadvantages

- Household RO units use a lot of water because they have low back pressure
- As a result they recover only 5 to 15 percentage of the water entering the system
- The remainder is discharged as wastewater
- An RO unit delivering 5 gallons of treated water per day may discharge 40 to 90 gallons of wastewater per day to the septic system
- Large scale industries/ municipal system have a production efficiency of closer to 48% because they can generate the high pressure needed for RO filtration

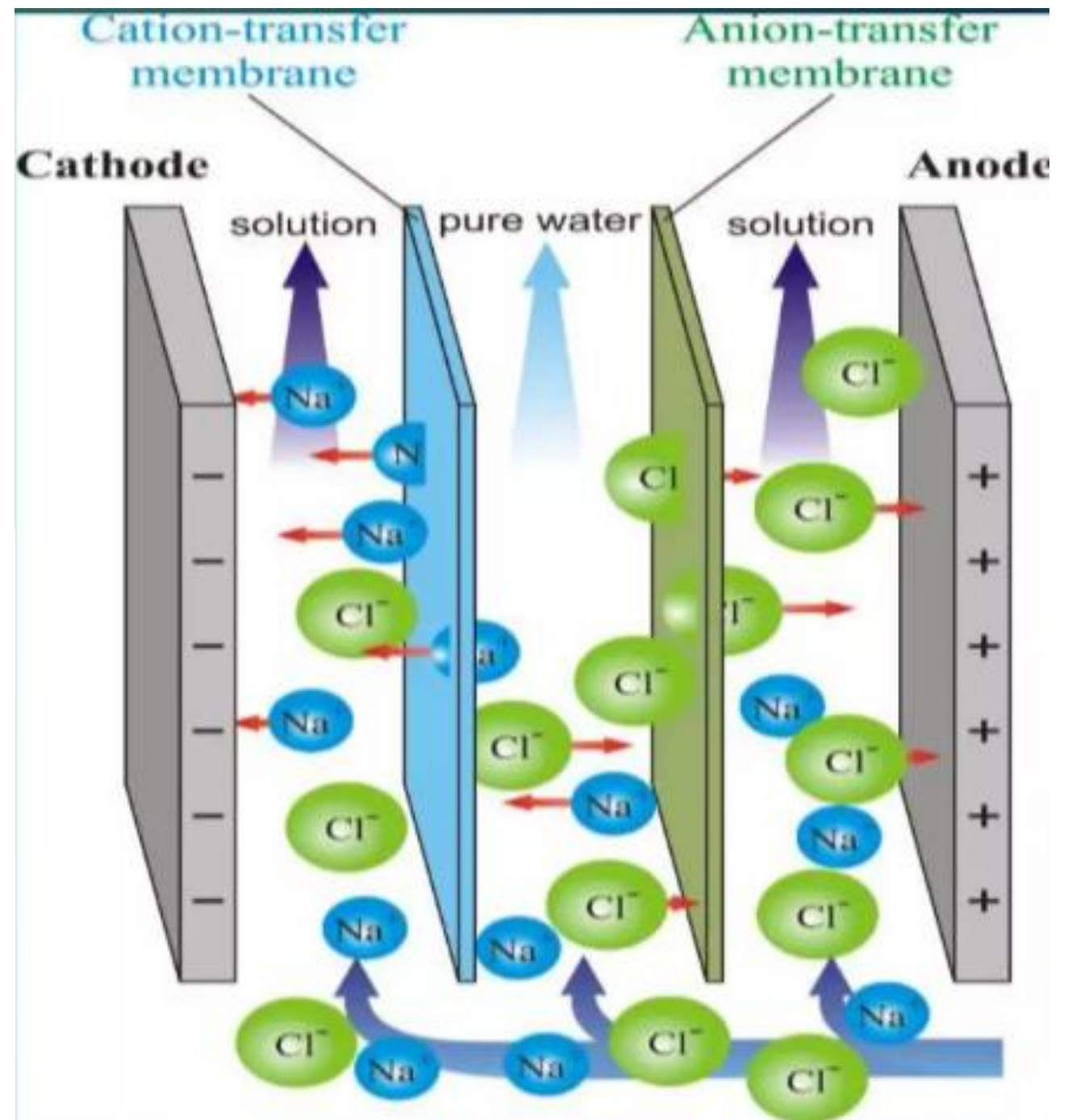
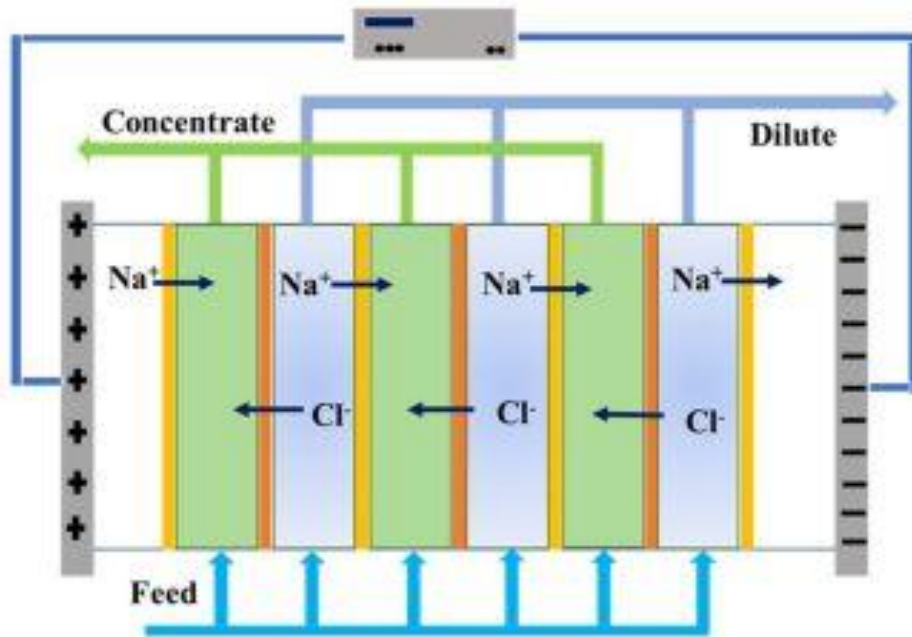
Electrodialysis

Introduction

- Dialysis
 - A membrane process where solutes diffuse from one side of the membrane (feed side) to the other (permeate side) according to their concentration gradient
- Electrodialysis
 - Ion exchange membrane separation process in which electrical potential used as the driving force for the separation
- EDR
 - Polarity of the electrodes is reversed, so the permeate becomes the retentate and vice versa

Components

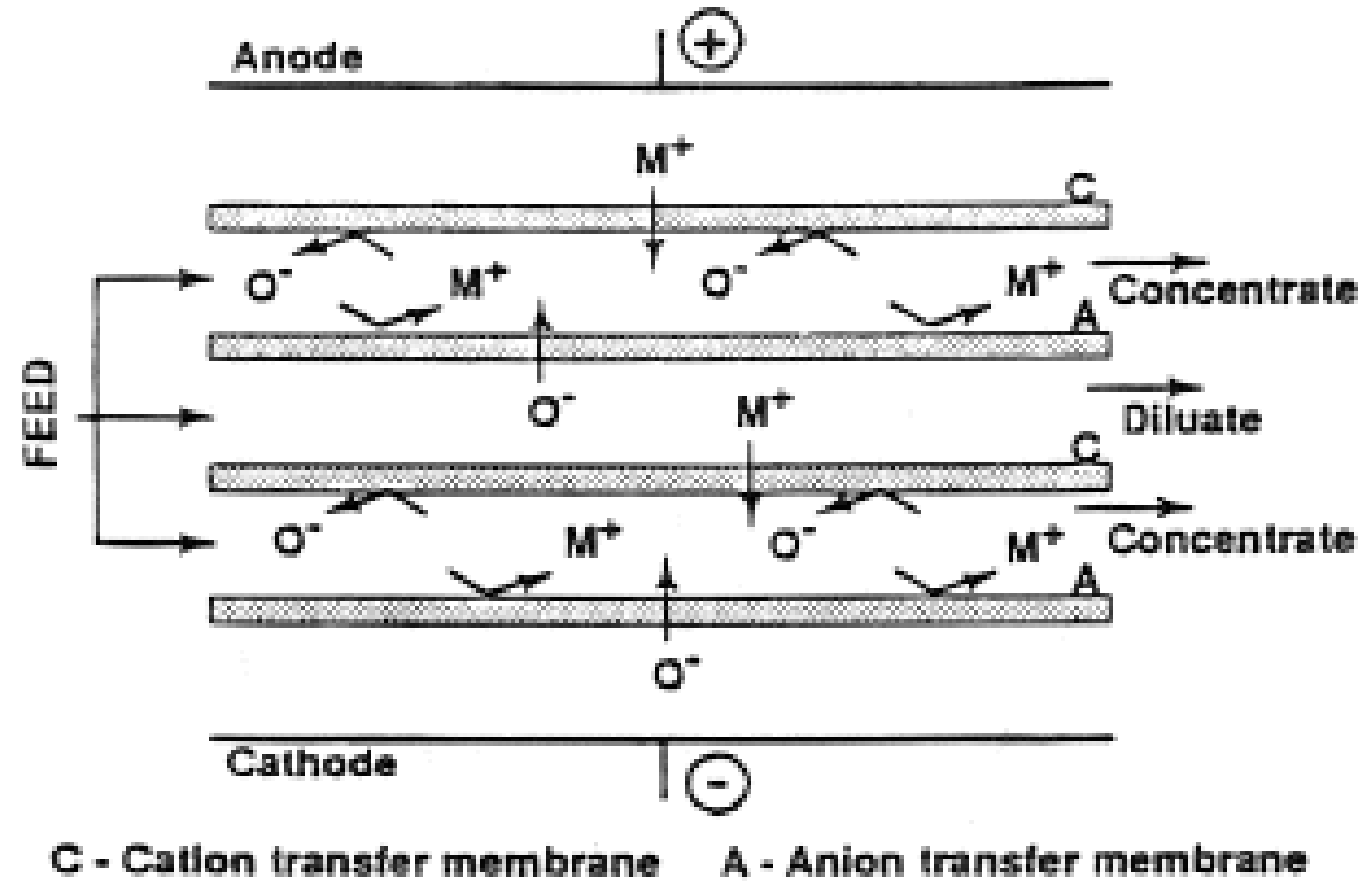
- Power source
- Cathode and Anode
- Ion exchange membrane



Process

- Electrodialysis – process for the separation of electrolyte from a solvent, typically water
- Widely used in the desalination of water and process solutions
- It uses a direct electrical current to transport ions through sheets of ion-exchanger membranes and is operated in a unit with at least three compartments
- The terminal compartments house an anode and a cathode, between which a potential difference is applied to drive the ions through the electrolyte solutions and the membranes

Process contd...



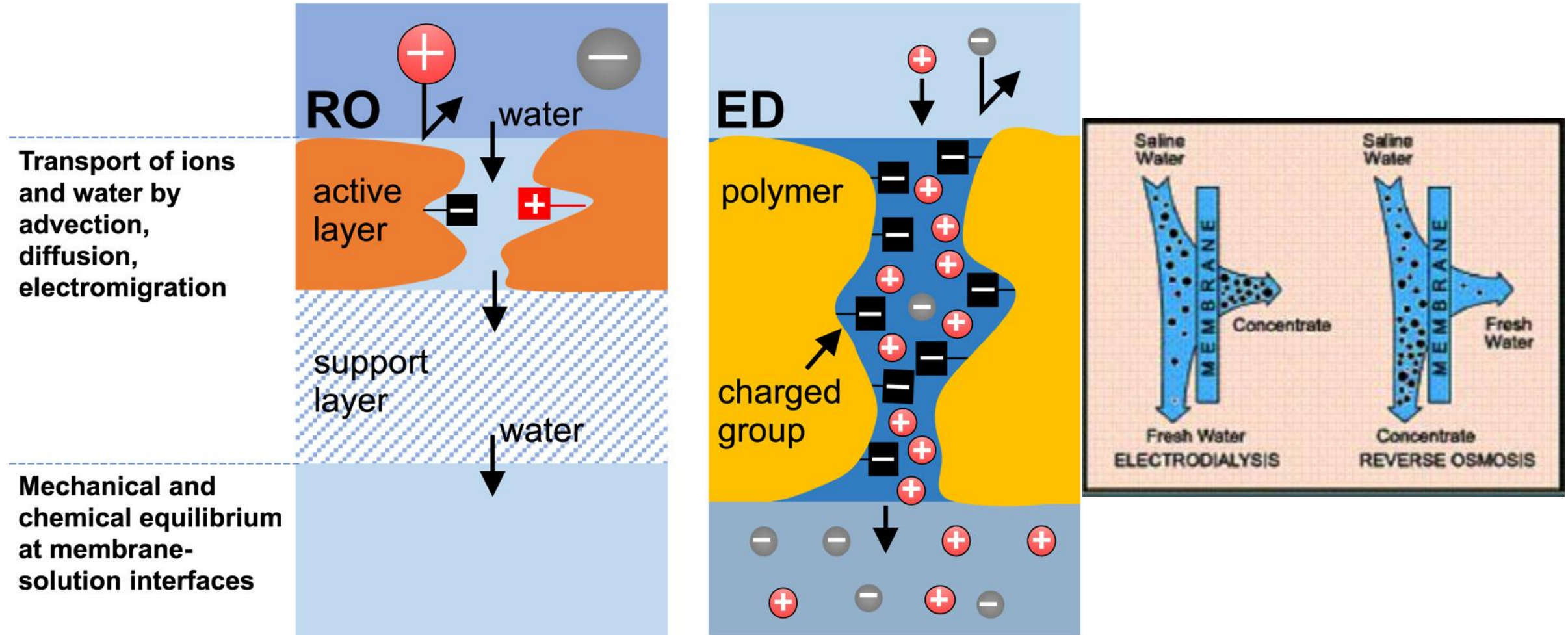
Process contd....

- Two types of membrane are used: one which is preferentially permeable to the transport anions (anion selective), and one which is preferentially permeable to cations (cation-selective)
- Membranes are arranged alternately between the electrodes, forming individual compartments (or cells)
- The solution to be desalinated is held in one compartment and during current flow , anions move through the anion exchange membrane in the direction of the anode into adjacent compartment while cations move in the opposite direction into an adjacent compartment on the otherside

Process contd....

- The solution becomes depleted in ions in one compartment and solutions in adjacent compartments become enriched in ions
- In practice, solutions flow through the compartment to allow continuous operation and several hundred cell pairs(one concentrated and one diluted solution) are used

RO vs ED



ED vs RO contd....

- **Electrodialysis (ED)**

- ✓ ED is a voltage driven process
- ✓ Electrodialysis is usually applied to deionisation of aqueous solutions
- ✓ Sometimes pre-treatment is necessary
- ✓ High conversion ratio (nearly 80%)

- **Reverse Osmosis (RO)**

- ✓ RO is a pressure-driven process
- ✓ RO is used to remove suspended solids, total organic carbon (TOC), or other contaminants
- ✓ Pretreatment is necessary always
- ✓ Low conversion ratio (b/w 30 to 80%)

Advantages and disadvantages of ED

• Advantages

- ✓ Low energy consumption
- ✓ High conversion ratio (nearly 80%)
- ✓ Low space and material requirements
- ✓ The product water needs only limited pre-treatment
- ✓ Higher brine concentration achievable

• Disadvantages

- ✓ Only removes ions, organic and colloids removed
- ✓ Selection of membranes and stacks and highly dependent on feed water chemistry
- ✓ Purity affected by quality of feed water

Applications

- Desalination of brackish water
- Boiler feedwater production
- Waste and process water treatment
- Table salt production
- Desalination of contaminated industrial water for reuse
- Reduce electrolyte content from effluent streams

Assignment Questions

- Explain the working of
 - Rotating Biological Reactors
 - Sequencing Batch Reactors